AstroTracker

Step #1

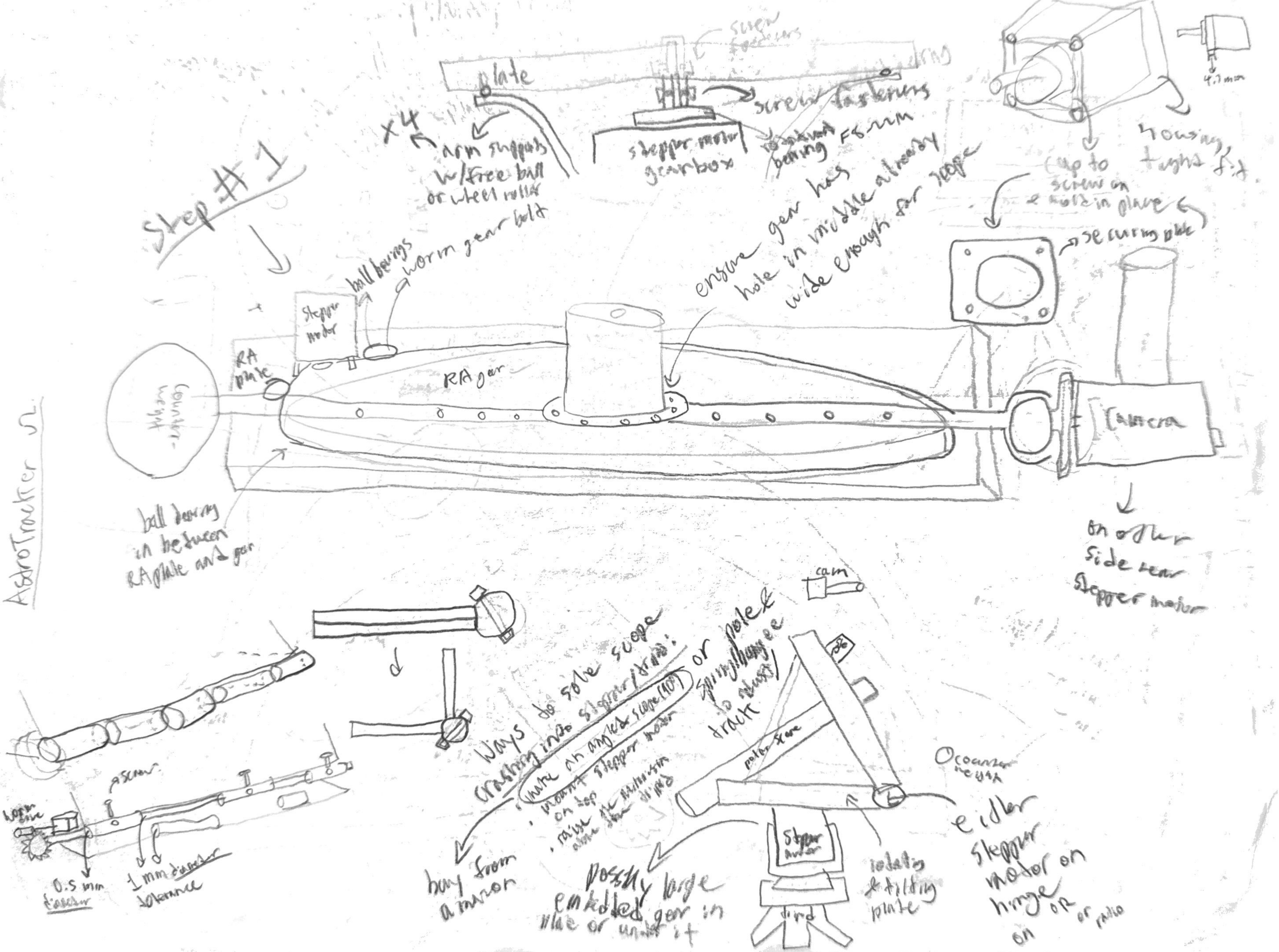
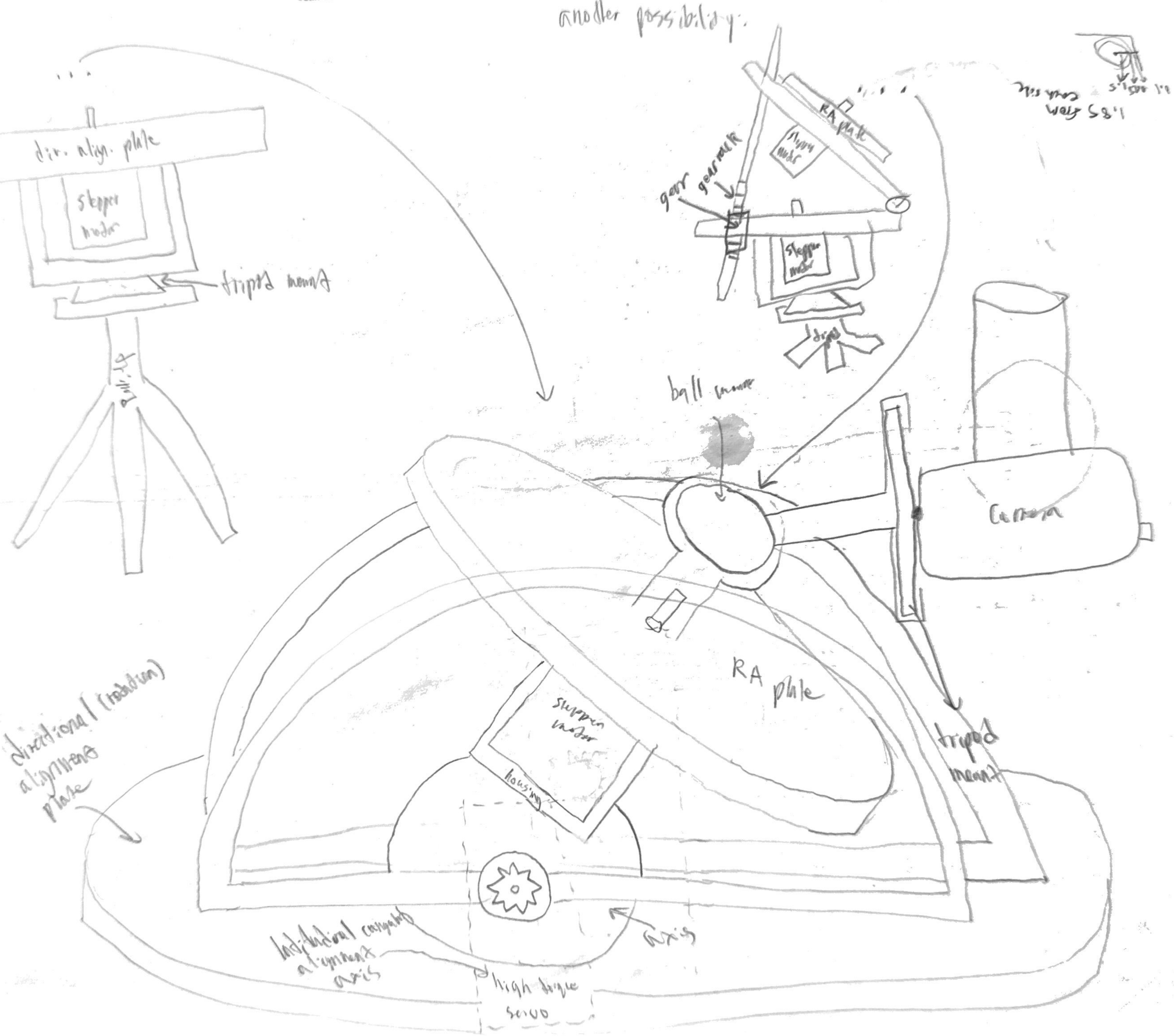


plate
x4
arm supports w/free ball or wheel roller
screw fasteners
stepper motor gearbox
bearing F8-22M
screw fasteners
4.7 mm
housing, tight fit
cap to screw on & hold in place
securing plate
ensure gear has hole in middle already wide enough for scope
ball bearings
worm gear bolt
stepper motor
RA plate
RA gear
counter-weight
camera
on other side near stepper motor
ball bearing in between RA plate and gear
cam
Ways to solve scope crashing into stepper/drive:
make an angled scope (40°)
, mount stepper motor on top
, raise the [illegible] above [illegible]
or pole & spring/hinge to adjust track
polar scope
counter weight
idler
stepper motor on hinge on or or radio
rotating & tilting plate
stepper motor
tripod
possibly large embedded gear in plate or under it
buy from amazon
worm drive
screw
0.5 mm diameter
1 mm diameter tolerance

AstroTracker v1.



$C = \pi d$ $\qquad \frac{\#\text{teeth}}{\text{pitch}} = C = \pi d$

$d = \frac{200(2)}{\pi} = \frac{400}{\pi} \Rightarrow d = \frac{TP}{\pi}$

127.324 mm

0.9° or 1.8° stepper → can always switch out motor

1:12.5 physical ratio 16T : 200T

1/64 microstepping or 1/54 → can always switch

~8.1 arcsecs/microstep (1.8°)

Adequate torque ✓

Target ratio = 1:100

1/16 microstepping

16T gear → x T gear

16 : 1600

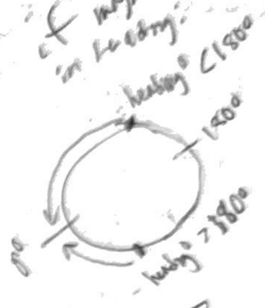
×

1/16 microstepping

16 : 1000

100T gear

if my preference for return 'in heading:



530 mm GT2 belt
~65T

20T toothless idler bearing

400 mm C-C dist

28.8 degs/s = 360 degs/s
0.00418 degs/s = x 0.05223 degs/s

$\frac{360\text{ degs}}{28.8\text{ degs}} = \frac{x\text{ degs}}{0.00418\text{ degs}}$

$x = 0.05223$ degs

$\frac{0.05223\ \cancel{\text{degs}}/s}{1} \cdot \frac{12800\text{ microsteps/s}}{360\ \cancel{\text{degs}}/s} = 1.85692$ microsteps/s
↳ 7% error

16T : 220T gear ratio incr.???
↳ 2.044 microsteps/s
↳ only 2% error

16T : 216T even closer?
↳ 2.005 mics/s
↳ 0.25% error

check compatibility w/ gear belt lengths

10000000.0ms

100000 s

1666 2/3 min

27 7/9 hr

1.1574 day

↓

3 7/9 hr

↓

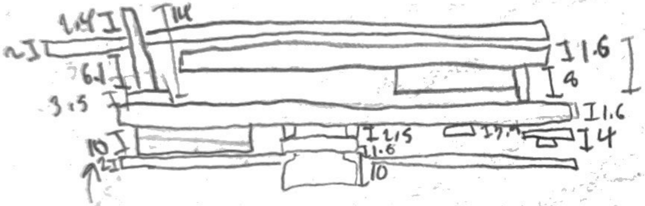
46 2/3 min

↓

40.05

↓

0.0ms

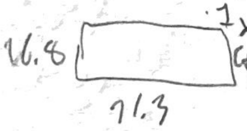


NEB:
- M2.5 nut and bolt x30mm
- M3 nut and bolt x16mm   13.2
- M3 nut and bolt x6mm

Total w/ boards: 2 + 9.6 + 1.6 + 10 + 2 = 25.2 mm
w/ 10: 21.2 mm

Spacers:
- 3x M2.5, offset 1mm, 9.6mm L spacer
- 1x same but 0.75 offset
- 4x M3, offset 1mm, 8mm L spacer
- 4x M2.5, offset 1mm, 10mm L spacer
- 1x M4, offset 1mm, 0.8mm L spacer
- 1x M4, (for top) offset 3.3-9.6=1.35mm 5mm L spacer (ALREADY (ADDED))

-40   -37.64



. hexagonal design on PCB case for heat dissipation?

Kdam Thmor Pro

Kenia

Jayden is Awsome!